Recall: $\pi: X \to Y = \operatorname{Spec} A$, qc. sep.

$\forall\, 0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ ses in $\operatorname{Qcoh}(X)$,

$$0 \to \underset{\substack{\| \\ H^0(Y, \pi_* \mathcal{F}')}}{H^0(X, \mathcal{F}')} \to H^0(X, \mathcal{F}) \to \cdots$$

$$0 \to \pi_* \mathcal{F}' \to \pi_* \mathcal{F} \to \pi_* \mathcal{F}'' \to \widetilde{H^1(X, \mathcal{F}')} \to \widetilde{H^1(X, \mathcal{F})} \to \cdots$$

When Y not affine:

Def: (Higher pushforward sheaves):

$\pi: X \to Y$ qc. sep, then

$\forall\, i \geq 0$, $\exists\, R^i \pi_*: \operatorname{Qcoh}(X) \to \operatorname{Qcoh}(Y)$, s.t

① $R^0 \pi_* = \pi_*$

② $\forall\, 0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ ses in $\operatorname{Qcoh}(X)$,

$\exists$ les $0 \to R^0 \pi_* \mathcal{F}' \to \cdots$

③ $\forall\, U: \operatorname{Spec} A \hookrightarrow Y$, $R^i \pi_* \mathcal{F}(U) = H^i_A(\pi^{-1}(U), \mathcal{F}|_{\pi^{-1}(U)})$.

Then $\forall \mathcal{F} \in Qcoh(X)$, $R^i\pi_*\mathcal{F}$ is called the ith higher pushforward sheaf of $\mathcal{F}$

Well-defineness: Object: Show $\forall\, U = \operatorname{Spec} A \hookrightarrow Y$,

$\forall f \in A$, $\forall \mathcal{F} \in Qcoh(X)$, $U_f := \operatorname{Spec} A_f \hookrightarrow U$.

$$\begin{tikzcd}
H^i_A(\pi^{-1}(U), \mathcal{F}|_{\pi^{-1}(U)}) \arrow[r] \arrow[dr, "\mathrm{Res}"'] \arrow[dr, phantom, shift right=3ex, "\circlearrowright"] & H^i_{A_f}(\pi^{-1}(U_f), \mathcal{F}|_{\pi^{-1}(U)}) \arrow[d, dashed, "\wr"] \\
& H^i_A(\pi^{-1}(U), \mathcal{F}|_{\pi^{-1}(U)}) \otimes_A A_f
\end{tikzcd}$$

The Čech complex of $\mathcal{F}|_{\pi^{-1}(U)}$:

$$0 \to C^0 \to C^1 \to \cdots$$

$\otimes_A A_f$ + cohomology: $H^i_{A_f}(\pi^{-1}(U_f), \mathcal{F}|_{\pi^{-1}(U_f)})$.

cohomology + $\otimes_A A_f$: $H^i_A(\pi^{-1}(U), \mathcal{F}|_{\pi^{-1}(U)}) \otimes_A A_f$.

$\otimes_A A_f$ exact: ✓.

$I \rightsquigarrow \exists\, R^i\pi_*: Qcoh(X) \to Qcoh(Y)$ satisfies ①. ③.

Morphism: $\forall\, \mathcal{F}' \to \mathcal{F}$, $\forall\, U = \operatorname{Spec} A \hookrightarrow Y$, $\forall f \in A$

Čech complex: $C^\bullet(\mathcal{F}'|_{\pi^{-1}(U)}) \to C^\bullet(\mathcal{F}|_{\pi^{-1}(U)})$

$\otimes_A A_f$ + cohomology: $H^i_{A_f}(\pi^{-1}(U_f), \mathcal{F}'|_{\pi^{-1}(U_f)}) \to H^i_{A_f}(\pi^{-1}(U_f), \mathcal{F}|_{\pi^{-1}(U_f)})$

cohomology + $\otimes_A A_f$: $H^i_A(\pi^{-1}(U), \mathcal{F}'|_{\pi^{-1}(U)}) \otimes_A A_f \to H^i_A(\pi^{-1}(U), \mathcal{F}|_{\pi^{-1}(U)}) \otimes_A A_f$.

$\forall$ ses $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$.

Čech complex: $0 \to C^\bullet(\mathcal{F}'|_{\pi^{-1}(U)}) \to C^\bullet(\mathcal{F}|_{\pi^{-1}(U)}) \to C^\bullet(\mathcal{F}''|_{\pi^{-1}(U)}) \to 0$

$\otimes_A A_f$ + cohomology: —— ——

cohomology + $\otimes_A A_f$: —— —— —— ——

✓.

Rem: ① $R^i\pi_*$ is the ith right derived functor of the left exact functor π_*.

② For general $U \hookrightarrow Y$, $R^i\pi_*\mathcal{F}(U) = H^i(\pi^{-1}(U), \mathcal{F})$ is not true.

③ Grothendieck's Coherence thm: $\pi: X \to Y$ proper, Y locally Noe, $\mathcal{F} \in Coh(X)$, then $\forall i \geq 0$, $R^i\pi_*\mathcal{F} \in Coh(Y)$

(pf when π is projective: Thm 18.1.3).

18.7.A (easy): When π is affine, $\forall i > 0$, $R^i\pi_* = 0$.

pf: Locally: $\forall U = \operatorname{Spec} A \hookrightarrow Y$, $\mathcal{F} \in Qcoh(X)$,

$$R^i\pi_*\mathcal{F}(U) = H^i_A(\pi^{-1}(U), \mathcal{F}) = 0, \ \forall i > 0.$$

18.7.B. (base change).

(a) $\psi: Y' \to Y$. $\pi: X \to Y$. π qc + sep, $\mathcal{F} \in Qcoh(X)$.

$X' = X \times_Y Y'$.

$$\begin{tikzcd} X' \arrow[r, "\psi'"] \arrow[d, "\pi'"'] \arrow[dr, phantom, "\ulcorner"] & X \arrow[d, "\pi"] \\ Y' \arrow[r, "\psi"] & Y \end{tikzcd}$$

Then $\exists$ natural morphism $\psi^*(R^i\pi_*\mathcal{F}) \to R^i\pi'_*(\psi')^*\mathcal{F}$

(b) $\psi': Y' \to Y$ aff. $\exists$ open cover $\{\operatorname{Spec} A_i\}_i$ of Y,

s.t. $\psi^{-1}(\operatorname{Spec} A_i) = \operatorname{Spec} B_i$, B_i flat A_i-alg.

(i.e. $\otimes_{A_i} B_i$ exact).

Then $\psi^*(R^i\pi_*)\mathcal{F} \to R^i\pi'_*(\psi')^*\mathcal{F}$ isom.

pf: (a) Wlog let $Y = \operatorname{Spec} A$, $Y' = \operatorname{Spec} B$.

$R^i\pi_*\mathcal{F} \rightsquigarrow$ Čech complex $C^\cdot$

$$\psi^*(R^i\pi_*\mathcal{F}) = \overbrace{H^i(C^\cdot)\otimes_A B}$$

$$R^i\pi'_*(\psi')^*\mathcal{F} = \overbrace{H^i(C^\cdot \otimes_A B)}$$

$\otimes_A B$ right exact: $H^i(C^\cdot)\otimes_A B \to H^i(C^\cdot \otimes_A B)$.

$\rightsquigarrow \psi^*(R^i\pi_*)\mathcal{F} \to R^i\pi'_*(\psi')^*\mathcal{F}$.

(b) $\otimes_{A_i} B_i$ exact ✓. $\square$.

18.7.C.

$$\begin{tikzcd} X' \arrow[r, "\psi'"] \arrow[d, "\pi'"'] \arrow[dr, phantom, "\circlearrowright"] & X \arrow[d, "\pi"] \\ Y' \arrow[r, "\psi"] & Y \end{tikzcd}$$

π, π' qc. sep, $\Rightarrow \forall \mathcal{F} \in \mathrm{Qcoh}(X)$, $\exists$ natural morphism

$$\psi^*(R^i\pi_*\mathcal{F}) \longrightarrow R^i\pi'_*(\psi')^*\mathcal{F}$$

Pf:

$$\begin{tikzcd} X' \arrow[r, "\psi_1'"] \arrow[d, "\pi'"'] \arrow[dr, phantom, "\circlearrowright"] & X \times_Y Y' \arrow[r, "\psi_2'"] \arrow[d, "\pi_{Y'}"'] \arrow[dr, phantom, "\circlearrowright"] & X \arrow[d, "\pi"] \\ Y' \arrow[r, equal] & Y' \arrow[r, "\psi"] & Y \end{tikzcd}$$

By 18.7.(B) (a),

$$\psi^*(R^i\pi_*\mathcal{F}) \to R^i(\pi_{Y'})_*(\psi_2')^*\mathcal{F}.$$

So wlog let $Y = Y' = \mathrm{Spec}\, A$

$$R^i\pi_*\mathcal{F} = R^i(\pi')_*(\psi')^*\mathcal{F}.$$

$H^i(X, \mathcal{F}) \to H^i(X', (\psi')^*\mathcal{F})$. (P494. (iv)) $\square$

e.g (18.7.17) $q \in Y$,

$$\begin{tikzcd} \pi^{-1}(q) \arrow[r, "\psi'"] \arrow[d, "\pi'"] \arrow[dr, phantom, "\ulcorner"] & X \arrow[d, "\pi"] \\ \operatorname{Spec} k(q) \arrow[r, "\psi"] & Y \end{tikzcd}$$

$$(R^i\pi_*\mathcal{F}) \otimes k(q) = \psi^* R^i\pi_*\mathcal{F} \to R^i(\pi')_*(\psi')^*\mathcal{F}$$
$$\parallel$$
$$H^i(\pi^{-1}(q), \mathcal{F}|_{\pi^{-1}(q)}).$$

18.7.E. (Projection formula)

$\pi: X \to Y$ qc, sep, $\mathcal{F} \in \operatorname{Qcoh}(X)$, $\mathcal{G} \in \operatorname{Qcoh}(Y)$.

(a) $(R^i\pi_*\mathcal{F}) \otimes \mathcal{G} \longrightarrow R^i\pi_*(\mathcal{F} \otimes \pi^*\mathcal{G})$.

(b) isom when $\mathcal{G}$ locally free

pf: Wlog let $Y = \operatorname{Spec} A$.

$\mathcal{F} \rightsquigarrow$ Čech complex $C^\cdot$

$\otimes G$ right exact

$$\rightsquigarrow (R^i\pi_*\mathcal{F})\otimes G = H^i(C^\cdot)\otimes G \longrightarrow H^i(C^\cdot\otimes\pi^*G)$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \| $$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad R^i\pi_*(\mathcal{F}\otimes\pi^*G)$$

(a) ✓.

(b): G locally free $\Rightarrow$ $\otimes G$ exact. $\square$

18.7.5 Thm: $\pi: X\to Y$ proj. Y locally Noe,

$m = \max\{\dim\pi^{-1}(q): q\in Y\}$. then $\forall i>n$, $R^n\pi_*=0$.

pf: wlog let $Y=\operatorname{Spec}A$.

$\forall p\in Y$. $\dim\pi^{-1}(p)\le m$. $k:=k(p)$.

$$\begin{tikzcd}
\pi^{-1}(p) \arrow[r, hook] \arrow[d] & \mathbb{P}^n_k \arrow[r] \arrow[d] & \operatorname{Spec}k \arrow[d] \\
X \arrow[r, hook] & \mathbb{P}^n_A \arrow[r] & Y=\operatorname{Spec}A
\end{tikzcd}$$

By ex 12.3.D, $\exists$ hypersurface $V(f_1),\dots,V(f_{m+1})$ in $\mathbb{P}^n_k$

s.t $\pi^{-1}(p)\cap\left(\bigcap_{i=1}^{m+1}V(f_i)\right)=\phi$, i.e $\pi^{-1}(p)\subseteq\bigcup_{i=1}^{m+1}D(f_i)$

$f_1, \cdots, f_{m+1} \in \Gamma(\mathbb{P}^n_k, \mathcal{O}(1)) \hookrightarrow k[x_0, \cdots, x_n]$.

$\Big\downarrow$ lift

$\tilde{f}_1, \cdots, \tilde{f}_{m+1} \in A_p[x_0, \cdots, x_n]$

$\exists f \in A - p$ s.t. $\tilde{f}_1, \cdots, \tilde{f}_{m+1} \in A_f[x_0, \cdots, x_n]$, $p \in D(f)$.

$$\begin{tikzcd}
\pi^{-1}(D) \arrow[r, hook] \arrow[d] & \mathbb{P}^n_k \arrow[r] \arrow[d] & \operatorname{Spec} k \arrow[d] \\
X_{A_f} \arrow[r, hook] \arrow[d] & \mathbb{P}^n_{A_f} \arrow[r] \arrow[d, "\phi"] & D(f) = \operatorname{Spec} A_f \arrow[d, "\phi"] \\
X \arrow[r, hook] & \mathbb{P}^n_A \arrow[r] & Y = \operatorname{Spec} A
\end{tikzcd}$$

$Z = X \cap \left(\bigcap_{i=1}^{m+1} V(\tilde{f}_i) \right) \hookrightarrow \mathbb{P}^n_{A_f}$.

$$\begin{tikzcd}
Z \arrow[r] \arrow[d, hook] & X \arrow[r, "\pi"] \arrow[d, hook] & Y \\
\mathbb{P}^n_{A_f} \arrow[r, "cl"] & D(f) \arrow[ur] &
\end{tikzcd}$$

$\rightsquigarrow W = \operatorname{Im}(Z \to Y) \hookrightarrow D(f)$

$p \notin W$

$(\pi^{-1}(p) \cap Z = \pi^{-1}(p) \cap \left(\bigcap_{i=1}^{m+1} V(\tilde{f}_i) \right) = \phi)$

$\exists$ affine open $\operatorname{Spec} B \hookrightarrow Y$ s.t. $p \in \operatorname{Spec} B \subseteq D(f)$,

, $\operatorname{Spec} B \cap W = \phi$.

$\tilde{f}_i \in A_f[x_0, \cdots, x_n]$

$\downarrow$

$\tilde{f}_i' \in B[x_0, \dots, x_n]$.

$\phi = \pi^{-1}(\operatorname{Spec} B) \cap Z = \pi^{-1}(\operatorname{Spec} B) \cap \left(\bigcap_{i=1}^{m+1} V(\tilde{f}_i')\right)$.

$\Rightarrow \pi^{-1}(\operatorname{Spec} B) = \bigcup_{i=1}^{m+1} D(\tilde{f}_i')$.

$\Rightarrow \forall q \geq m+1, \mathcal{F} \in \operatorname{Qcoh}(X), R^q\pi_*\mathcal{F}|_{\operatorname{Spec} B} = 0$

$\Rightarrow R^q\pi_*\mathcal{F} = 0$. $\square$

18.7.F: $\pi: X \to Y$ proper. X, Y Noe, $\mathcal{L}$ π-ample line bundle on X. Then $\forall \mathcal{F} \in \operatorname{Coh}(X)$, $\forall m \gg 0$, $R^i\pi_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) = 0$.

Recall: $\mathcal{L}$ π-ample: $\forall \operatorname{Spec} B \hookrightarrow Y$, $\mathcal{L}|_{\pi^{-1}(\operatorname{Spec} B)}$ ample over B.

pf: Y Noe $\Rightarrow$ Wlog let $Y = \operatorname{Spec} A$

Thm 18.1.3 (Serre vanishing) ✓. $\square$

18.8.

Thm: A Noe ring, X proper over A, $\mathcal{L}$ line bundle on X

Then $\mathcal{L}$ ample on $X \Leftrightarrow \forall \mathcal{F} \in \mathrm{Coh}(X)$, $\exists n_0$ s.t $\forall n \geq n_0$,

$H^i(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n}) = 0$, $\forall i > 0$.

Thm: X qc, sep, Then TFAE

(a) X aff

(b) $\forall \mathcal{F} \in \mathrm{Qcoh}(X)$, $H^i(X, \mathcal{F}) = 0$, $\forall i > 0$

(c) $\forall \mathcal{F} \in \mathrm{Qcoh}(X)$, $H^1(X, \mathcal{F}) = 0$.

18.9. Grothendieck's Coherence thm: $\pi: X \to Y$ proper, X, Y locally Noe, then $\forall \mathcal{F} \in \mathrm{Coh}(X)$, $\forall i \geq 0$, $R^i \pi_* \mathcal{F} \in \mathrm{Coh}(Y)$

Chow's lem: $\pi: X \to \mathrm{Spec} A$ proper, A Noe.

Then $\exists \mu: X' \to X$ surj and proj, s.t $\pi \circ \mu$ proj, and μ is an isom on a dense open subset of X.

$$\begin{tikzcd} X' \arrow[rr, "\mu"] \arrow[dr, "\pi \circ \mu"'] & & X \arrow[dl, "\pi"] \\ & \mathrm{Spec} A. & \end{tikzcd}$$

(proper = proj - birational proj)